Problem

We consider a graph, G with edges E and vertices V, which is divided into subgraphs, $G_1, \ldots, G_i, \ldots, G_k \subset G$, where each subgraph is given to one of k players. Each player knows only their own graph. The goal is, given some starting vertex v, and some end vertex, w to:

1. Identify if it is possible to construct a sequence of vertices $v, v_1, \ldots, v_i, w$ such that $\forall (m, n) \in i, m \neq n : (v_m, v_n) \in \bigcup_{i=0}^{k} E_i$, where E_i is the set of edges belonging to G_i. In other words, find if there is some sequence of vertices who's edges are entirely within at least one of $G_1 \ldots G_k$, which connects the vertices v and w.

2. If such a path exists, find the minimum number of edges needed to get from v to w, while using only edges belonging to $\bigcup_{i=0}^{k} E_i$

Lower bound sketches and ideas

For item 1, we think this would require computing $\binom{k}{2}$ pairwise set disjointnes in the worst case, to determine where the intersections of each G_i are, at which point each player could on their own determine the possible paths between each of the subgraphs that intersect with their own.

For item 2, we would be required to compute $\binom{k}{2}$ pairwise intersections, since in order to minimize the path we would need to know all possible paths that could connect v and w.